Udayasree Punuru
COMETFWC067

Detailed Connections Between Arduino, 7447 IC and Seven Segment Display

Aim

To interface a Seven Segment Display with Arduino Uno using 7447 BCD to Seven Segment Decoder IC and display digits 0-9.

1. Components Used

Component
Arduino Uno
Seven Segment Display (Common Anode)
7447 BCD to Seven Segment Decoder IC
Jumper Wires
Breadboard

2. Purpose of Each Component

Component	Function
Arduino Uno	Sends BCD (Binary Coded Decimal) output to 7447 IC
7447 IC	Decodes BCD input and drives seven segment display segments
Seven Segment Display	Displays decimal digits 0-9
Jumper Wires	Connects Arduino pins to 7447 IC and display
Breadboard	Holds components and connections

3. Power Connections (7447 IC)

Pin	Connection
Pin 16 (Vcc)	Arduino +5V
Pin 8 (GND)	Arduino GND
Control Pins 3, 4, 5	+5V

4. Arduino to 7447 IC Connections

Arduino Pin	7447 IC Pin	Signal
D2	Pin 7 (A)	BCD Bit A (LSB)
D3	Pin 1 (B)	BCD Bit B
D4	Pin 2 (C)	BCD Bit C
D5	Pin 6 (D)	BCD Bit D (MSB)

5. 7447 IC to Seven Segment Display Connections

7447 IC Pin	Seven Segment Pin	Segment
Pin 13 (a-bar)	Pin 7	a (Top)
Pin 12 (b-bar)	Pin 6	b (Top Right)
Pin 11 (c-bar)	Pin 4	c (Bottom Right)
Pin 10 (d-bar)	Pin 2	d (Bottom)
Pin 9 (e-bar)	Pin 1	e (Bottom Left)
Pin 15 (f-bar)	Pin 9	f (Top Left)
Pin 14 (g-bar)	Pin 10	g (Middle)

6. Seven Segment Display Common Pins

Pin	Connection
Pin 3 (COM)	Arduino +5V
Pin 8 (COM)	Arduino +5V

7. Seven Segment Display Pin Description

Pin No.	Segment	Position on Display
1	e	Bottom Left vertical
2	d	Bottom horizontal
3 & 8	COM	Common Anode - +5V
4	c	Bottom Right vertical
5	dp	Decimal Point
6	b	Top Right vertical
7	a	Top horizontal
9	f	Top Left vertical
10	g	Middle horizontal

8. Overall Working

1. Arduino sends 4-bit BCD value through pins D2-D5 to 7447 IC.
2. 7447 IC decodes the BCD input and activates the corresponding segment output pins.
3. Since Common Anode display: segment turns ON when 7447 output is LOW.
4. Seven Segment Display shows the corresponding decimal digit (0-9).
5. Common Anode pins (3 & 8) of Seven Segment are connected to +5V.

9. Data Flow of the System

Arduino Output Pins (D2, D3, D4, D5)
↓
7447 BCD to Seven Segment Decoder IC
↓
Seven Segment Display Segments (a - g)
↓
Displayed Decimal Digit (0 - 9)

10. Important Notes

* Use a Common Anode Seven Segment Display only.
* Common Anode pins (3 & 8) must be connected to +5V.
* 7447 IC Vcc (Pin 16) to +5V and GND (Pin 8) to GND.
* Control pins 3, 4, 5 of 7447 must be connected to +5V.
* 7447 IC outputs are active LOW - segment turns ON when output is LOW.
* Ensure common GND between Arduino, 7447 IC, and display.
* cd = is decoder / seven segment use (noted in diagram).